

PART II (30 Marks)

Question I (8 marks)

Given a string T of length n , represented as $T[0]$ through $T[n-1]$, and another string P of length m , represented as $P[0]$ through $P[m-1]$, we say that P is a substring of T if there exists at least one index i ($0 \leq i \leq n-m$) such that the segment $T[i]$ through $T[i+m-1]$ matches $P[0]$ through $P[m-1]$.

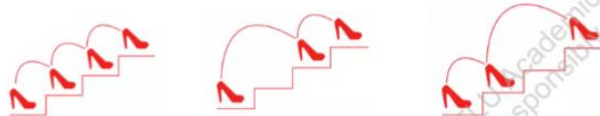
For instance, if $T = \text{"ACGTACGGG"}$, then "ACGG" is a substring that appears exactly once in T , "ACG" is a substring that appears twice, and "ACC" is not a substring of T at all.

Ensure your algorithm is clear and concise, and that your explanation of the time complexity accurately reflects the computational resources required by your pseudocode.

1. Write pseudocode for an algorithm that determines if the string P appears as a substring in the string T exactly once. 6
2. Determine the worst-case time complexity of your algorithm in Big O notation. Provide a brief explanation for your analysis. 2

Question II (18 marks)

Stairs-climbing problem: there are n stairs, a person standing at the bottom wants to climb stairs to reach the n -th stair. The person can climb either 1 stair or 2 stairs at a time, the task is to count the number of ways that a person can reach at the top.



Example: As shown in the figure above, starting from the bottom. To reach to the top, we have several options: 1) taking three steps individually, 2) taking two initially and then one, or 3) taking one step and then two.

1. Briefly describe the idea of the dynamic programming technique. 2
2. Let $F(n)$ be the number of ways that a person can reach the n -th stairs. For convenience, define $F(0)=0$, $F(1)=1$, $F(2)=1$. Set up a recurrence relation for $F(n)$ ($n > 0$) that can be used by a dynamic programming algorithm. 6
3. For the amount $n=6$, solving the stairs-climbing problem using the relation set in a) 4

n	0	1	2	3	4	5	6
F(n)	0	1	1				

4. Write pseudocode of the dynamic programming algorithm for solving this problem and determine its time complexity. **6**

Question III (4 marks)

Given any two decision problems A and B , what is a polynomial time reduction from A to B ? **4**
Briefly explain how this technique can be used to prove certain problems are NP- hard

PART II

Question 1 (10 marks)

1. Describe what a decision problem is and what an optimisation problem is. **3**
2. Describe what a P problem is and what a NP problem is. **3**
3. For each problem below, state whether it is a P problem or NP-complete problem.
 1. Vertex Cover Problem. **1**
 2. Finding minimum spanning tree (MST) in a weighted undirected graph **1**
 3. 0/1 Knapsack problem. **1**
 4. Finding the nth Fibonacci number. **1**

Question 2 (20 marks)

Consider the problem of searching for genes in DNA sequences using Horspool's algorithm. A DNA sequence is represented by a text on the alphabet $\{A, C, G, T\}$, and the gene or a gene segment is a pattern.

1. Construct the shift table for the following gene segment. **10**

TCCTATTCTT

2. Apply Horspool's algorithm to locate the pattern in the following DNA sequence. **10**

TTATAGATCTGGTATTCTTTTATAGATCTCCTATTCTT